

Medvale Internal Medicine

Cardiovascular Medicine - Chapter 1B

Chronic Coronary Syndrome / Stable Angina

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1. Why this matters clinically

Stable angina is the predictable form of coronary ischemia: symptoms appear when myocardial oxygen demand rises and resolve when demand falls. The major clinical task is to separate stable exertional symptoms from acute coronary syndrome, then decide whether the patient needs risk-factor treatment, antianginal therapy, noninvasive testing, coronary angiography, or revascularization.

This chapter intentionally includes numeric anchors because cardiology questions often test thresholds: symptom duration, EF cutoffs, stress-test heart-rate targets, stenosis percentages, LDL targets, medication doses, and revascularization clues.

2. Core illness script

Category	Stable angina script
Mechanism	Fixed atherosclerotic coronary narrowing limits blood-flow reserve during increased myocardial demand.
Typical trigger	Exertion, emotional stress, cold exposure, tachycardia, or a heavy meal.
Typical relief	Rest or sublingual nitroglycerin.
Typical duration	Usually 1-5 minutes; generally less than 10-15 minutes.
Troponin	Normal in uncomplicated stable angina.
Resting ECG	Often normal; may show prior MI, LVH, conduction disease, or nonspecific abnormalities.
Danger transition	New onset, worsening, rest pain, prolonged pain, or lower exertional threshold = possible ACS.

3. Numbers, cutoffs, and timelines

Topic	Number to know	Why it matters
Typical stable angina duration	1-5 min; generally <10-15 min	Longer, rest, or progressive pain suggests ACS rather than stable angina.
Male CAD age-risk threshold	>45 yr	Age increases pretest probability.
Female CAD age-risk threshold	>55 yr	Age increases pretest probability.
Premature CAD in male first-degree relative	<55 yr	Raises inherited/family-risk concern.
Premature CAD in female first-degree relative	<65 yr	Raises inherited/family-risk concern.
Normal EF anchor	>50%	More reassuring LV function.
Reduced EF prognostic anchor	<50%	Associated with higher mortality risk in CAD.
Left main coronary artery	Supplies about two-thirds of myocardium	Left main disease is high risk.
Exercise stress target	80-85% of predicted max HR	Needed for an adequate exercise stress test.
Predicted max HR	220 - age	Used to estimate stress-test target HR.

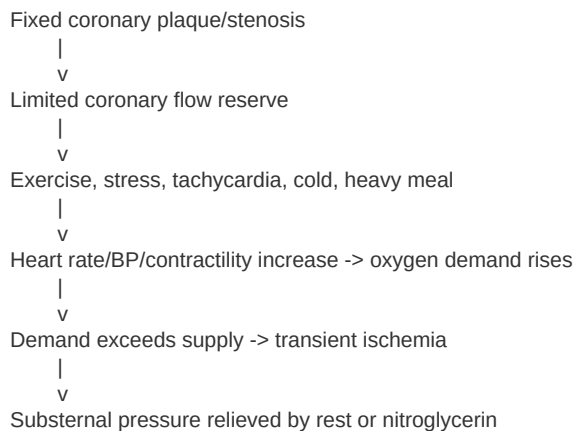
Topic	Number to know	Why it matters
Exercise ECG sensitivity anchor	~75% if adequate exercise and interpretable ECG	Useful but not perfect; false negatives occur.
Hemodynamically significant stenosis	Often >70% luminal narrowing	Can produce exertional angina.
Classic LDL target in known CAD	<70 mg/dL	Older high-yield target; modern care also emphasizes maximally tolerated statin and add-on therapy when indicated.
Dietary saturated fat goal	<7% total calories	Classic risk-reduction target.
Dietary cholesterol goal	<200 mg/day	Classic risk-reduction target.
Bare-metal stent restenosis anchor	~40% within 6 mo historically	Explains why drug-eluting stents reduce restenosis risk.

4. Pathophysiology: supply-demand mismatch

At rest, a partially narrowed coronary artery may still deliver enough blood. During exertion, heart rate, blood pressure, and contractility rise; the myocardium needs more oxygen. A normal coronary artery increases flow by vasodilating. A diseased artery distal to a fixed stenosis may already be maximally dilated, so it cannot increase flow enough. Transient ischemia produces pressure-like discomfort, then resolves when demand falls.

Key contrast: stable angina is usually fixed-demand ischemia. ACS is usually plaque rupture or erosion with thrombosis and abruptly reduced supply.

Original diagram: demand exceeds supply



5. CAD risk factors and prognostic anatomy

Risk/prognostic factor	High-yield detail
Diabetes mellitus	Major CAD risk factor; symptoms may be less classic. Multivessel CAD with diabetes often favors CABG when anatomy is suitable.
LDL elevation / hyperlipidemia	Drives plaque formation; high-intensity statin therapy is central.
Hypertension	Common CAD risk factor; increases myocardial workload.
Smoking	Promotes endothelial dysfunction, thrombosis, and plaque instability.
Family history	Premature CAD: male first-degree relative <55 yr or female first-degree relative <65 yr.
LV systolic function	EF >50% is more reassuring; EF <50% worsens prognosis.
Coronary distribution	Left main, proximal LAD, and two-/three-vessel disease increase risk.

6. Presentation patterns

Pattern	Definition	Clinical interpretation
Typical angina	Substernal discomfort + provoked by exertion/stress + relieved by rest/nitro	Highest clinical fit for stable angina when pattern is predictable.
Atypical angina	Two of the three classic features	Can still be CAD, especially with risk factors.
Nonanginal pain	Zero or one classic feature	Consider noncardiac causes, but do not ignore high-risk features.
Unstable pattern	Rest pain, new severe pain, increasing frequency/duration/intensity, or less exertion required	Treat as possible ACS, not routine outpatient stable angina.

Clinical anchor: Stable angina generally lasts minutes. Pain lasting only seconds is less typical for ischemia; pain lasting many hours or occurring at rest should raise concern for ACS or a nonanginal diagnosis, depending on the rest of the presentation.

7. Stable angina vs ACS

Feature	Stable angina	Unstable angina / NSTEMI / STEMI
Usual trigger	Predictable exertion or stress	Rest, minimal exertion, new onset, or worsening pattern
Pathology	Fixed flow-limiting stenosis	Plaque rupture/erosion with thrombosis
Duration	Often 1-5 min; generally <10-15 min	Often prolonged or recurrent
ECG	Often normal at rest	May show ST depression, T-wave inversion, or ST elevation
Troponin	Normal	Normal in UA; elevated in NSTEMI/STEMI
Site of care	Usually outpatient if stable and low risk	ED/inpatient monitored evaluation

8. Diagnostic approach

Step 1 - Do not stress-test unstable symptoms

If the patient is having active concerning pain, rest pain, hemodynamic instability, syncope, ischemic ECG changes, or a rising troponin, manage as possible ACS. Stress testing is for stable patients after immediate danger has been excluded.

Step 2 - Estimate pretest probability

Pretest probability is driven by age, sex, symptom quality, and risk factors. Low-probability patients may not need immediate cardiac testing; intermediate-probability patients often need noninvasive testing; high-probability patients may need medical therapy plus imaging or invasive evaluation if testing will change management.

Step 3 - Choose an anatomic or functional test

Clinical question	Testing category	Examples
Is there coronary plaque/stenosis?	Anatomic testing	Coronary CT angiography; invasive coronary angiography
Is there inducible ischemia?	Functional testing	Exercise ECG, stress echo, nuclear perfusion imaging, stress cardiac MRI
How much myocardium is at risk?	Stress imaging	Stress echo, nuclear perfusion imaging, stress CMR

9. Stress test selection with numbers

Test	Best use	Numeric/high-yield anchor	Avoid or limitations
Exercise ECG	Can exercise + interpretable baseline ECG	Adequate test aims for 80-85% predicted max HR; predicted max HR = 220 - age; sensitivity anchor ~75% when adequate	Avoid if LBBB, paced rhythm, baseline ST changes, LVH repolarization abnormalities, or cannot exercise.

Test	Best use	Numeric/high-yield anchor	Avoid or limitations
Stress echocardiography	Need ischemia assessment when ECG is not interpretable; also assesses valves/LV function	Ischemia appears as inducible wall-motion abnormality	Less reliable with poor acoustic windows or extensive baseline wall-motion abnormalities.
Nuclear perfusion imaging	Need perfusion map or ischemic burden	Reversible defect = ischemia; fixed defect = scar/infarct	Radiation; can be less helpful in some baseline conduction patterns such as LBBB depending on protocol.
Pharmacologic stress	Patient cannot exercise adequately	Vasodilator stress: adenosine/dipyridamole/regadenoson; demand stress: dobutamine	Agent choice depends on asthma/COPD, AV block, caffeine intake, BP, and local protocol.

Worked numeric example

A 60-year-old patient has predicted maximal heart rate = $220 - 60 = 160$ beats/min. An adequate exercise stress test generally aims for about 80-85% of that value: approximately 128-136 beats/min. If the patient stops far below this range because of fatigue or orthopedic limitation, the test may be nondiagnostic.

10. Interpreting stress tests

Finding	Meaning	Next decision
ST-segment depression during exercise	Subendocardial ischemia pattern	Positive exercise ECG; consider CAD severity and need for imaging/cath.
Chest pain, hypotension, or significant arrhythmia during stress	Higher-risk positive test	Stop test, stabilize, and escalate evaluation.
Reversible perfusion defect	Stress hypoperfusion with rest recovery = inducible ischemia	May be salvageable myocardium; consider medical optimization and/or revascularization depending on burden.
Fixed perfusion defect	Defect persists at stress and rest = scar/prior infarct	Revascularization less likely to restore dead tissue.
Inducible wall-motion abnormality	Ischemic myocardium contracts abnormally during stress	Positive stress echo.

11. Coronary angiography and stenosis thresholds

Coronary angiography is the definitive anatomic test for coronary stenosis and can be paired with PCI when appropriate. It is usually reserved for patients with high-risk noninvasive testing, refractory symptoms despite medical therapy, suspected high-risk anatomy, reduced EF thought to be ischemic, or diagnostic uncertainty when the result will change management.

Angiographic concept	Number/anchor	Clinical use
Significant coronary stenosis	Often >70% luminal narrowing	Can limit flow enough to produce angina.
Left main disease	Large myocardial territory; classic high-risk anatomy	Often favors CABG if anatomy and patient factors permit.
Two-/three-vessel CAD	Worse prognosis than single-vessel disease	More likely to need surgical discussion, especially with diabetes or reduced EF.
Positive stress test followed by cath	Cath clarifies anatomy and revascularization suitability	PCI can be done during diagnostic procedure when indicated.

12. Treatment goals

Stable angina treatment has two goals: prevent cardiovascular events and relieve symptoms. These overlap but are not identical. Aspirin and lipid lowering are event-prevention therapies. Nitrates primarily relieve symptoms. Beta-blockers can relieve symptoms and are especially important after MI or in selected high-risk CAD patients.

Event-prevention therapy

Therapy	Role	Numbers/cautions
Smoking cessation	Major risk-reduction intervention	Risk reduction begins quickly; classic teaching notes substantial CAD risk reduction by 1 year after quitting.
Statin therapy	Plaque stabilization and LDL reduction	Classic known-CAD LDL target <70 mg/dL; modern practice uses high-intensity or maximally tolerated statin plus add-ons when needed.
Antiplatelet therapy	Reduces thrombotic events in appropriate CAD patients	Use only when benefit exceeds bleeding risk.
BP control	Reduces vascular events and myocardial workload	Avoid excessive hypotension that worsens ischemia.
Diabetes management	Reduces vascular risk; selected agents provide CV benefit	SGLT2 inhibitors/GLP-1 receptor agonists may be favored in appropriate patients.
Diet	Improves cardiometabolic risk	Classic targets: saturated fat <7% calories; cholesterol <200 mg/day.

Antianginal symptom therapy

Medication	Mechanism	Use	High-yield cautions
Beta-blockers	Decrease HR, BP, and contractility -> lower oxygen demand	First-line for many exertional-angina patients	Avoid severe bradycardia, high-grade AV block, and acute decompensated HF.
Nitrates	Venodilation lowers preload; coronary vasodilation may help supply	Acute relief and pre-exertion prophylaxis	Avoid with PDE-5 inhibitors; hypotension, headache, tolerance.
Calcium channel blockers	Vasodilation; non-DHP agents also lower HR/contractility	Add-on therapy; preferred for vasospastic angina	Avoid non-DHP CCBs in reduced EF.
Ranolazine	Reduces late sodium current; less HR/BP effect	Persistent angina when HR/BP limits other meds	QT prolongation and drug interactions.

Nitroglycerin safety anchor: Ask about phosphodiesterase-5 inhibitor use before nitrates. Combining nitrates with sildenafil/tadalafil/vardenafil can cause dangerous hypotension.

13. Revascularization decisions: medical therapy, PCI, or CABG

Revascularization is not automatic in stable CAD. Many stable patients are managed first with guideline-directed medical therapy and risk-factor modification. Revascularization is most compelling for unacceptable symptoms despite medical therapy or anatomy associated with high risk.

Choice	Best fit	Key number/anatomy anchors
Medical therapy alone	Mild symptoms, low-risk testing, symptoms controlled	Continue risk-factor treatment and antianginal therapy.
PCI	Focal culprit lesions causing symptoms despite medications	Excellent for symptom relief; restenosis risk historically higher with bare-metal stents (~40% within 6 mo), lower with drug-eluting stents.
CABG	Left main disease, complex multivessel CAD, diabetic multivessel CAD, or selected reduced EF patients	Left main and three-vessel patterns are high-risk; CABG often favored when anatomy is suitable.

Original decision map

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Stable angina diagnosis likely
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Start/optimize medical therapy + risk-factor control
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+--- Symptoms controlled and low-risk testing -> continue medical therapy
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- +-- Symptoms persist despite therapy -> coronary angiography
 - +-- Focal lesion suitable for PCI -> PCI for symptom relief
 - +-- Left main / complex multivessel / diabetic multivessel -> CABG discussion
- +-- High-risk stress test or reduced EF suspected ischemic -> define anatomy

14. Differentials and mimics

Condition	How it mimics angina	Clue that separates it
GERD	Burning substernal discomfort	Postprandial, sour taste, worse lying down.
Esophageal spasm	Squeezing pain; may respond to nitrates	Dysphagia or meal association; can be hard to distinguish clinically.
Costochondritis	Anterior chest pain	Reproducible focal tenderness.
Aortic stenosis	Exertional chest pain	Systolic crescendo-decrescendo murmur, syncope, dyspnea.
Hypertrophic cardiomyopathy	Exertional pain or syncope	Murmur increases with Valsalva; family history.
Microvascular angina	Angina with nonobstructive coronaries	Ischemia symptoms/tests but no major epicardial stenosis.
Variant angina	Rest pain, often at night	Transient ST elevation during pain; responds to CCBs/nitrates.

15. High-yield exam clues

Stem clue	Most likely answer
Substernal pressure with exertion, relieved by rest in 3 min	Stable angina.
Pain now occurring at rest or increasing in frequency/duration	Unstable angina/ACS pathway.
Can exercise + normal resting ECG	Exercise ECG is reasonable.
LBBB, paced rhythm, or baseline ST changes	Stress imaging rather than plain exercise ECG.
Stress defect that resolves at rest	Reversible ischemia.
Stress defect that persists at rest	Scar/prior infarct.
EF <50% with CAD	Worse prognosis; consider ischemic evaluation and GDMT.
Left main or three-vessel CAD	High-risk anatomy; CABG often considered.
Rest chest pain at night + transient ST elevation	Variant angina.
Nitrate prescribed + PDE-5 inhibitor use	Dangerous hypotension risk.

16. Common pitfalls

- **Calling changing angina stable.** New, worsening, rest, or prolonged pain is ACS until proven otherwise.
- **Using exercise ECG when the ECG cannot be interpreted.** LBBB, paced rhythm, LVH repolarization changes, or baseline ST depression often require imaging.
- **Forgetting the heart-rate target.** A patient who cannot reach about 80-85% predicted max HR may have a nondiagnostic exercise test.
- **Thinking PCI is always mortality-preventing in stable angina.** PCI is often symptom-improving; event prevention still depends on medical therapy and risk-factor control.
- **Ignoring diabetes in anatomy decisions.** Diabetes plus multivessel CAD often pushes decision-making toward CABG discussion.
- **Giving nitrates without checking PDE-5 inhibitors or blood pressure.** Severe hypotension can result.

17. Practice questions

Question 1

A 58-year-old man develops substernal pressure after walking uphill. It resolves after 4 minutes of rest. His resting ECG is normal. What is the most likely diagnosis?

Answer: Stable angina. **Teaching point:** Predictable exertional substernal discomfort relieved within minutes by rest is the classic pattern.

Question 2

A 60-year-old patient is scheduled for an exercise ECG. What heart-rate range roughly indicates an adequate test?

Answer: Predicted max HR = $220 - 60 = 160$. Adequate stress is about 80-85%, or roughly 128-136 beats/min.

Question 3

A 65-year-old woman with exertional chest pain can exercise, but her baseline ECG shows LBBB. What type of testing is preferred over plain exercise ECG?

Answer: Stress imaging. **Teaching point:** LBBB makes ST-segment interpretation unreliable.

Question 4

A stress nuclear perfusion study shows an inferior wall defect during stress that disappears at rest. What does this mean?

Answer: Reversible ischemia. **Teaching point:** The myocardium is underperfused with stress but viable at rest.

Question 5

A patient with stable angina has left main coronary disease on angiography. Why is this high risk?

Answer: The left main supplies a large myocardial territory, classically about two-thirds of the heart, so obstruction threatens a large amount of myocardium.

Question 6

A patient has stable angina despite beta-blocker and nitrate therapy. Coronary angiography shows complex three-vessel CAD and diabetes. What revascularization strategy is commonly favored if anatomy and surgical risk are acceptable?

Answer: CABG. **Teaching point:** Diabetes with multivessel CAD often favors surgical revascularization over PCI.

18. Chapter summary

Stable angina is predictable exertional ischemia caused by demand exceeding supply in chronic coronary disease. Numeric anchors matter: symptoms usually last 1-5 minutes and generally less than 10-15 minutes; exercise testing aims for 80-85% of predicted max HR, where predicted max HR is $220 - \text{age}$; EF below 50% worsens prognosis; stenoses greater than about 70% can be flow limiting; and left main or multivessel CAD is high-risk anatomy. Management combines risk-factor control, event-prevention therapy, antianginal symptom therapy, and selective revascularization when symptoms persist or anatomy is high risk.

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